INCH-POUND

MIL-C-44358A
7 May 1991
SUPERSEDING
MIL-C-44358
9 November 1988

MILILTARY SPECIFICATION

COVER, INDIVIDUAL, CAMOUFLAGE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 60 by 96 inch, polyester, mesh knit cover intended to provide instant camouflage for the individual soldier.

1.2 Classification. The cover shall be of one type and size in the following classes as specified (see 6.2).

Class 1 - Woodland Camouflage Pattern
Class 2 - Daytime Desert Camouflage Pattern
Class 3 - Snow Camouflage (white)
Class 4 - Desert Camouflage Pattern (3 color)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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SPECIFICATIONS

FEDERAL

A-A-50083	-	Bags, Plastic, Folded Garment
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-1-1516	-	Woodland Pattern - 60 Inches
2-1-1884	-	Daytime Desert Pattern - 60 Inches
2-1-2240	-	Desert Camouflage Pattern (3 color) - 48 and 60 Inches

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UX, Natick, MA 01760-5017.)

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Application for copies should be addressed to the Executive Director, American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, Durham, NC 27709-2215.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Standard sample. The face and back of the dyed and/or printed mesh cloth shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Yarn. The yarns shall be semi-dull or dull polyester. Yarns for class 1, class 2 and class 4 covers shall be without optical brighteners.

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3.4 Color. The color of the finished cover shall be as follows:

Class 1 - Woodland Camouflage Pattern
Class 2 - Daytime Desert Camouflage Pattern
Class 3 - Snow Camouflage (white)
Class 4 - Desert Camouflage (3 color)

The required shades for classes 1, 2 and 4 shall be obtained by dying the ground shade with disperse dyes and wet overprinting with disperse dyes. Transfer printing and the use of resin bonded pigments are not permitted.

3.4.1 Matching. The color shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200K. The color of the class 1 and 2 camouflage covers shall be evaluated by placing a single layer sample mesh cloth and standard sample side by side on a backing of Dark Blue 150 Tropical worsted cloth. Dark Blue 150 may be obtained by contacting U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-ITCP, Natick, MA 01760.

3.4.2 Colorfastness. The finished cloth shall show fastness to light, perspiration, and laundering equal to or better than the standard sample or equal to or better than a rating of "good" for light and perspiration, and "fair" for laundering for each of the pattern areas. The finished cloth shall show fastness to crocking equal to or better than the standard sample or shall show an AATCC Chromatic Transference Scale rating not lower than 3.5 for each of the pattern areas. Testing shall be as specified in 4.4.4.

3.5 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.4.4.

TABLE I. Physical requirements

Characteristic	Requirement
Weight, oz./sq.yd.	1.5 to 1.8
Meshes per inch:	
Wales	4 to 8
Courses	5 to 8
Bursting strength, lb.	30 (min.)

3.5.1 Knit. The cloth shall be mesh type warp knit.

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3.5.2 Width. The width of the finished cover shall be 60 ± 1 inches, including selvage.

3.6 Finishing. The cloth shall be scoured and heat set. Class 1, 2 and 4 shall be dyed and/or printed to meet the requirements specified in 3.6.1.

3.6.1 Pattern execution. The pattern of the finished cloth shall reproduce the standard Woodland, Daytime Desert, or Desert (3 color) Camouflage pattern in respect to design, color placement, and registration of the respective areas. The warpwise pattern repeat shall be 27.25, +1.25 -3.5 inches for the class 1-Woodland Camouflage pattern, and 16.75, +1.25 -1.75 inches for the class 2-Daytime Desert Camouflage pattern and class 4-Desert Camouflage pattern. Each pattern area shall show solid coverage; skitteriness exceeding that shown by the standard sample in any of the printed areas will not be acceptable. NOTE: If haloing or off register of the print pattern does not exceed 1/16 inch, the pattern execution is acceptable. (Usually occurs on the reverse side of the print). When the standard sample is not referenced for pattern execution, a pattern drawing shall be provided and the pattern shall match that of the applicable drawing (see 2.1.2).

3.7 Curling. The finished cloth shall lie flat without distortion and shall show no evidence of curling when tested as specified in 4.4.4.

3.8 pH. The pH value of the water extract of the finished cloth shall be not less than 5.0, and not greater than 8.0 when tested as specified in 4.4.4.

3.9 Length. The finished cloth shall be cut in lengths of 96 ± 1 inches for each finished cover.

3.10 Workmanship. The end item shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality

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program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3 and tested for the characteristics specified in 4.4.4.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the yarn conforms to the requirements specified in 3.3.1.

4.4.2 End item visual examination. The end items shall be examined for the defects listed in table II. The defects found shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the cover, in which case, only one defect shall be counted. The lot size shall be expressed in units of covers. The sample unit shall be one cover. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5.

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TABLE II. End item visual defects

Examine	Defect
Knit	Hole, break, weak place, mend Thin area Loose or floated end Coarse yarn more than twice the thickness of normal yarn Broken or missing yarn Loose knitting, causing sleaziness
Color	Off shade
Cleanness	Objectionable odor Overall uncleanness
Pattern execution	Pattern execution not as specified

4.4.3 End item dimensional examination. The end items shall be examined for the measurement defects in table III. The lot size shall be expressed in units of one cover. The sample unit shall be one cover. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

TABLE III. Dimensional defects

Examine	Defect
Cover	Length of cover less than 95 inches or more than 97 inches Width of cover less than 59 inches or more than 61 inches, including selvage.

4.4.4 End item testing. The end items shall be tested for the characteristics listed in table IV. The methods of testing specified in FED-STD-191 wherever applicable and as listed in table IV shall be followed. The lot size shall be expressed in units of one cover. The sample unit shall be one cover. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. The sample size (number of sample units) shall be in accordance with the following:

Lot size (covers)	Sample size
800 or less	2
801 to 22,000	3
22,001 and up	5

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TABLE IV. End item tests

Characteristic	Requirement paragraph	Test method
Colorfastness to:		
Light (40 hours)	3.4.2	5660
Crocking	3.4.2	5651 1/
Laundering	3.4.2	5614
Perspiration	3.4.2	5680
Weight	3.5	5041
Meshes per inch	3.5	2/
Bursting strength	3.5	5120
Type of knit	3.5.1	3/
Curling	3.7	4.4.4.1
pH	3.8	2811

1/ Class 1, class 2 and class 4 - test single layer of mesh backed by white cloth.

2/ Five determinations per sample unit and the average reported to the nearest whole number.

3/ Unless otherwise specified, a certificate of compliance shall be submitted and shall be acceptable for the stated requirement.

4.4.4.1 Curling. Two specimens of cloth, 1-1/2 inches wide by 6 inches long shall be cut, one having the long dimension parallel to the wales and the other with the long dimension parallel to the courses. Both specimens shall be placed on a flat surface for at least 5 minutes and then visually examined for evidence of curling.

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flap, improper strapping, or inadequate stapling. Bulged or distorted container.
Content	Number of unit packs more or less than specified.

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Load not bonded as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each individual camouflage cover shall be folded to approximately 12 by 18 inches and shall be unit packed in a snug-fitting, flat style, clear polyethylene film bag conforming to A-A-50083.

5.1.2 Commercial. Each individual camouflage cover shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

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5.2.1 Level A packing. Forty-eight individual camouflage covers, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each fiberboard shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. The shipping container shall have approximate inside dimensions of 23-1/2 inches in length, 15-1/2 inches in width, and 10-1/2 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that inspection shall be in accordance with 4.4.5. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Forty-eight individual camouflage covers, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety DW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to CF, class domestic, variety SW, grade 275 of PPP-B-636. The shipping container shall have approximate inside dimensions of 23-1/2 inches in length, 15-1/2 inches in width, and 10-1/2 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that inspection shall be in accordance with 4.4.5.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance wit 4.4.5.

5.2.3 Commercial. Individual camouflage covers, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), individual camouflage covers, packed as specified in 5.2.2 or 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 3 in accordance with appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

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5.4.1 Polyethylene bagged unit packs. Polyethylene bagged unit packs shall have the required identification information listed below legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification. The bag or label shall only indicate the following information:

STOCK NUMBER
CONTRACT NUMBER
ITEM DESCRIPTION
CLASS - POLYESTER

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cover is intended to provide instant camouflage for the individual soldier.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Class required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article is required (see 3.1, 4.3, and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load (see 5.2.1)
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (See 5.2.2.1).
h. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to standard sample, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Subject term (key word) listing.

Cloth
Desert

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Mesh knit
Snow
Woodland

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Preparing activity:

Army - GL

(Project 8415-0814)

Review activities:

Army - MD
Air Force - 82
DLA - CT

User activity:

Air Force - 45

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-44358A	1991 May 7

3. DOCUMENT TITLE

COVER, INDIVIDUAL, CAMOUFLAGE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-4532	(2) AUTOVON 256-4532
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-IRT Natick, MA 01760-5019	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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